SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 Describe construction and working of Draft Tube crystallization with neat and clean diagram. (CO3)

Q.24 Describe the construction & working with a full labeled neat diagram for an Ion Exchange. (CO4)

Q.25 Derive the operating line for Rectification section in distillation. (CO1)

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Roll No. ....

5th Sem/ Chemical

Subject : Mass Transfer Operations-II

Time : 3 Hrs. M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

Q.1 Distillation is a process used to : (CO1)

a) Mix two liquids

b) Separate mixtures based on boiling point

c) Increase temperature

d) Change colour of a solution

Q.2 Azeotropic distillation is used to separate : (CO1)

a) Simple solutions

b) Immiscible liquids

c) Liquids that form constant boiling mixtures

d) Solids from liquids

Q.3 Leaching is used to extract material from : (CO2)

a) Gas b) Liquid

c) Solid d) None

Q.4 Crystallization is used to make : (CO3)

a) Pure solid b) Water

c) Oil d) Steam

Q.5 In adsorption, the material being adsorbed is called: (CO2)

a) Adsorbent b) Solute

c) Adsorbate d) Residue

Q.6 Feed plate divides the column into : (CO1)

a) Equal parts

b) Random parts

c) Rectifying & stripping sections

d) Heating & cooling zones

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The liquid collected after condensation in distillation is called_____. (CO1)

Q.8 Write any one application of leaching operation _____ (CO2)

Q.9 Expand the term U.F. (CO4)

Q.10 Define Adsorption. (CO4)

Q.11 Extraction is used in the chemical industry for separating _____ mixture. (CO2)

Q.12 The McCabe - Thiele method is a _________ method. (CO1)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Differentiate between Differential & Simple Distillation. (CO1)

Q.14 Describe the construction and working of a mixer-settler unit. Mention its advantages. (CO2)

Q.15 Write four advantage and disadvantage of adsorption as a separation technique. (CO3)

Q.16 Describe Reflux Ratio and Minimum Reflux ratio. (CO1)

Q.17 Explain the importance of bubble point and dew point in distillation column. (CO1)

Q.18 Describe about membrane separation. (CO1)

Q.19 Define Solubility and Super saturation with suitable examples. (CO3)

Q.20 Define Extraction. Explain the concept and application of Extraction. (CO2)

Q.21 Explain the key factors in selecting a solvent for extraction. (CO2)

Q.22 Explain the concept of Adsorption Isotherm. (CO4)